



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO SHRI MURUGAN AUTO FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

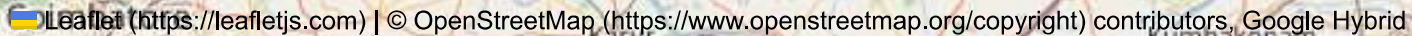
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



End: (12.131822, 78.390915)



# Welcome to the Journey Risk Management Study

## 1. Overview of the Route Map

The route from Sangagiri to Kosapatty in Tamil Nadu covers approximately 136.18 kilometers and takes about 2.69 hours under normal conditions for heavy vehicles carrying hazardous materials. The journey progresses through NH544 and various state highways, primarily in a rural setting, interspersed with small towns and villages.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate, with the following key weather patterns:

- **Dry Season (January to April):** Minimal rainfall, favorable for transportation.
- **Hot Season (May to August):** High temperatures can impact road conditions and driver performance.
- **Monsoon Season (September to December):** Heavy rains can cause waterlogging, slippery roads, and potential landslides, especially in hilly or poorly drained areas.

## 3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Typically during morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM).
- **Congestion Areas:** Potential bottlenecks include the urban centers you pass, such as Erode and smaller towns with narrow roads.
- **Avoidance Tips:** Plan travel outside peak hours or use bypasses where available.

## 4. Assessment of Road Quality and Infrastructure

- **Highways:** Generally well-maintained, but varying quality across state roads.
- **Bridges and Overpasses:** Inspect for weight limitations and structural integrity.
- **Potholes and Repairs:** Frequent in rural road sectors; maintenance inconsistency can affect travel speed and safety.

## 5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route 1:** Deviate through NH44 for a more direct route in severe weather or road blockage scenarios.
- **Alternative Route 2:** Local state roads connecting through inland paths provide detours avoiding major towns.

## 6. Local Regulations Affecting Hazardous Material Transport

- Permits required for transporting hazardous materials.
- Restrictions likely in residential and densely populated areas.
- Scheduled night travel might be mandated to lower risk.

## 7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

The region has a moderate history of incidents involving heavy vehicles due to rain-induced road damage or vehicle overturns during sharp turns in rural segments. Fatal incidents involving hazardous materials are rare but possible under poor weather or mechanical failures.

## 8. Environmental Considerations and Sensitive Areas

- **Protected Wildlife Zones:** Be aware of restrictions around reserved forests.
- **Agricultural Areas:** Watch for local farming traffic and livestock crossings.
- **Water Bodies:** Respect no-pollution zones near rivers and lakes, particularly during heavy rains.

## 9. Analysis of Communication Coverage

- **Good Coverage:** Along national highways and urban centers.
- **Potential Dead Zones:** Rural stretches, particularly between villages without cell towers.

## 10. Estimated Emergency Response Times

- **Urban Areas:** Erode or large towns - approximately 30-45 minutes.
- **Rural Areas:** Expect delays up to 1-2 hours depending on road conditions and remoteness.

## 12. Overall Risk Assessment Summary

The route presents moderate risks due to frequent weather changes, varying road quality, and potential traffic congestion in towns. However, with proper precautions, realistic timing, and adherence to local transport regulations, these risks can be mitigated. Notable care should be taken during monsoon seasons and on rural stretches where emergency services may have delayed reachability. Always ensure communication means remain operative and map alternate routes for any foreseeable obstructions.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 12.13658, 78.16171 | 15 KM/Hr    |
| 1  | Roundabout | High       | 12.14589, 78.17156 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4  | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 5  | Turn       | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 6  | Turn       | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 8  | Turn       | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 9  | Turn       | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 10 | Turn       | High       | 11.49224, 77.89601 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 11 | Turn      | Medium     | 11.53501, 77.95707 | 30 KM/Hr    |
| 12 | Turn      | Medium     | 11.53601, 77.95856 | 30 KM/Hr    |
| 13 | Turn      | Medium     | 11.62403, 78.10893 | 30 KM/Hr    |
| 14 | Turn      | High       | 11.64180, 78.11808 | 15 KM/Hr    |
| 15 | Turn      | High       | 11.64325, 78.11993 | 15 KM/Hr    |
| 16 | Turn      | Medium     | 11.69519, 78.10775 | 30 KM/Hr    |
| 17 | Turn      | Medium     | 11.73958, 78.05105 | 30 KM/Hr    |
| 18 | Turn      | Medium     | 11.75835, 78.05016 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.86847, 78.08367 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.87942, 78.08197 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.92585, 78.06507 | 30 KM/Hr    |
| 22 | Turn      | Medium     | 11.93704, 78.05402 | 30 KM/Hr    |
| 23 | Turn      | High       | 11.95818, 78.04978 | 15 KM/Hr    |
| 24 | Turn      | Medium     | 11.95988, 78.04948 | 30 KM/Hr    |
| 25 | Turn      | Medium     | 11.96304, 78.05136 | 30 KM/Hr    |
| 26 | Turn      | High       | 11.96399, 78.05135 | 15 KM/Hr    |
| 27 | Turn      | High       | 11.96580, 78.04923 | 15 KM/Hr    |
| 28 | Turn      | Medium     | 11.96675, 78.04939 | 30 KM/Hr    |
| 29 | Turn      | Medium     | 11.96791, 78.05070 | 30 KM/Hr    |
| 30 | Turn      | Medium     | 12.14026, 78.13955 | 30 KM/Hr    |
| 31 | Turn      | Medium     | 12.14048, 78.13998 | 30 KM/Hr    |
| 32 | Turn      | High       | 12.14499, 78.16500 | 15 KM/Hr    |
| 33 | Turn      | Medium     | 12.14721, 78.26926 | 30 KM/Hr    |
| 34 | Turn      | Medium     | 12.14553, 78.27731 | 30 KM/Hr    |
| 35 | Turn      | Medium     | 12.14597, 78.27877 | 30 KM/Hr    |
| 36 | Turn      | Medium     | 12.14467, 78.28372 | 30 KM/Hr    |
| 37 | Turn      | High       | 12.13069, 78.31362 | 15 KM/Hr    |
| 38 | Turn      | Medium     | 12.13248, 78.31918 | 30 KM/Hr    |
| 39 | Turn      | Medium     | 12.12880, 78.33678 | 30 KM/Hr    |
| 40 | Turn      | Medium     | 12.12942, 78.34092 | 30 KM/Hr    |
| 41 | Turn      | High       | 12.12751, 78.35960 | 15 KM/Hr    |
| 42 | Turn      | Medium     | 12.12857, 78.36048 | 30 KM/Hr    |
| 43 | Turn      | Medium     | 12.12645, 78.39367 | 30 KM/Hr    |
| 44 | Turn      | High       | 12.12746, 78.39480 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 0  | hospital | Thilagam Hospital                                               | 11.4753573,<br>77.8685829 | 30 km/h     | Medium     |
| 1  | clinic   | Meyyazhagan Clinic                                              | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |
| 2  | hospital | Dr. Jaganathan Hospital                                         | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3  | hospital | Kathir Eye Hospital                                             | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |
| 4  | police   | Police Quarters                                                 | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 10 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 12 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 19 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 20 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 21 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 22 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 24 | pharmacy | Healthy Pharmacy                   | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 25 | pharmacy | Thulasi Pharmacy                   | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 26 | hospital | Sri Gokulam Speciality Hospital    | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 27 | hospital | Global Medical Center              | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |
| 28 | hospital | Manipal Hospital                   | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |
| 31 | clinic   | Manimuthu Skin Clinic              | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 32 | hospital | St. Joseph's Speciality Hospital   | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 33 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 34 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 35 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 37 | clinic   | Bee Tee Clinic                     | 12.138889,<br>78.1576748  | 30 km/h     | Medium     |
| 38 | clinic   | Sheela Clinic                      | 12.138725,<br>78.1573436  | 30 km/h     | Medium     |
| 39 | clinic   | Suguna Clinic                      | 12.1386752,<br>78.1575468 | 30 km/h     | Medium     |
| 40 | hospital | Navina Hospital                    | 12.1367521,<br>78.1610296 | 30 km/h     | Medium     |
| 41 | clinic   | Bharani Neurologist                | 12.135969,<br>78.16255    | 30 km/h     | Medium     |
| 42 | hospital | D.N.V Hospital                     | 12.1351328,<br>78.162869  | 30 km/h     | Medium     |
| 43 | hospital | Pragadeesh Hospital                | 12.135284,<br>78.1591867  | 30 km/h     | Medium     |
| 44 | clinic   | Ragavendra Physiotherapy Clinic    | 12.135569,<br>78.1635247  | 30 km/h     | Medium     |
| 45 | hospital | Kamatchi Hospital                  | 12.1350608,<br>78.1594841 | 30 km/h     | Medium     |
| 46 | clinic   | Malar Siddha Clinic                | 12.1350485,<br>78.159226  | 30 km/h     | Medium     |

|    | type     | name                  | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------|---------------------------|-------------|------------|
| 47 | hospital | Kamala Nehru Hospital | 12.13404,<br>78.1633476   | 30 km/h     | Medium     |
| 48 | clinic   | Govindaraj Clinic     | 12.1344914,<br>78.1642372 | 30 km/h     | Medium     |
| 49 | hospital | Sri Priya Hospital    | 12.135347,<br>78.164391   | 30 km/h     | Medium     |

### Crowded Spots

|    | type       | name                                                    | coordinates               | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|---------------------------|-------------|------------|
| 13 | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 14 | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 16 | school     | Dist Education Training                                 | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 17 | school     | DIET - Salem                                            | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 18 | school     | Diamond Rays Matric HR Sec School                       | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 23 | college    | Sona College                                            | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |
| 29 | university | Periyar University                                      | 11.7186758,<br>78.0771089 | 30 km/h     | Medium     |
| 30 | school     | Fathima Hr Sec School                                   | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 36 | college    | Laxminarayana Polytechnic College                       | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |
| 50 | school     | Marudham Matric Hr Sec School                           | 12.1251245,<br>78.3823077 | 30 km/h     | Medium     |

### Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.13658, 78.16171



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.14589, 78.17156



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44900, 77.87406



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46400, 77.87323



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46765, 77.87037



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47403, 77.86841



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49149, 77.89601



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49224, 77.89601



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53501, 77.95707



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53601, 77.95856



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.62403, 78.10893**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.64180, 78.11808**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.64325, 78.11993**



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.69519, 78.10775



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73958, 78.05105



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.75835, 78.05016



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86847, 78.08367



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87942, 78.08197



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.92585, 78.06507



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.93704, 78.05402



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95818, 78.04978



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95988, 78.04948



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96304, 78.05136



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96399, 78.05135



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96580, 78.04923



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.96675, 78.04939**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.96791, 78.05070**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 12.14026, 78.13955**



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.14048, 78.13998



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.14499, 78.16500



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.14721, 78.26926



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.14553, 78.27731



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.14597, 78.27877



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.14467, 78.28372



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.13069, 78.31362



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.13248, 78.31918



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12880, 78.33678



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12942, 78.34092



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.12751, 78.35960



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12857, 78.36048



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12645, 78.39367



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.12746, 78.39480